

DA Notification Orchestration - Additional Requirements Technical Plan

1. Source Requirements Summary (Addendum)

The addendum introduces new functional requirements on top of outbound notification orchestration:

1. Email/Chat/Message Capture

- Capture inbound emails/messages/chats
- Store raw content for audit/compliance

2. Email/Chat/Message Parsing

- Clean message body (remove signatures, previous thread, noise)
- Extract candidate identifier fields (candidate id/email), message id, and cleaned message

3. Intent Detection

- Classify inbound response intent:
 - `accept`
 - `reject`
 - `request`
 - `query`

4. Workflow Automation

- Trigger downstream action based on intent:
 - invoke client API hook, or
 - publish to message queue for downstream consumption

5. Functional Flow

- System sends communication
- Recipient replies
- Message captured
- Parsed and cleaned
- Intent detected
- Workflow triggered
- System updated

2. Current Application Capability/Status

2.1 Existing Strengths (Already in Code)

- Multi-tenant outbound notification APIs
- Tenant auth model (API key + tenant JWT), admin auth, tenant portal

- Channel orchestration with priority and fallback modes
- Template management (tenant/global, preview, clone, versioning)
- Delivery tracking and status storage
- Provider health/failover and retry framework
- Deduplication and idempotency controls
- Usage quotas and dashboard endpoints
- Inbound **delivery status webhooks** (SES/Twilio/Slack/FCM) for out-bound tracking

2.2 Requirement Traceability Matrix (Addendum vs Current)

Requirement	Current Status	Evidence in Current App	Gap Level
Capture inbound candidate replies (email/chat/text)	Not Implemented	Existing webhooks are delivery/status oriented, not inbound conversational capture	High
Persist raw inbound message for audit	Not Implemented	No inbound message entity/table for raw payload storage	High
Parse/clean inbound body	Not Implemented	No parser pipeline (signature/thread stripping)	High
Extract candidate/message metadata	Not Implemented	No structured extractor for candidate_id/email/message_id from inbound	High
Intent detection (accept/reject/request/query)	Not Implemented	No inbound intent classifier endpoint/service	High
Trigger downstream workflow (client API hook / MQ)	Partially Implemented (infra only)	Celery/Redis infrastructure exists for outbound jobs, but no inbound intent workflow dispatcher	High

Requirement	Current Status	Evidence in Current App	Gap Level
End-to-end inbound response flow	Not Implemented	No inbound orchestration router or state tracking	High
Auditability for inbound decision chain	Not Implemented	Audit tables exist mainly for outbound events; no inbound audit domain	Medium-High

3. Pending Requirements (Gaps)

3.1 Functional Gaps

1. Inbound ingestion layer is missing

- No standardized endpoints/adapters for inbound email/SMS/WhatsApp/chat replies
- No source normalization contract

2. Inbound persistence model missing

- No tables for raw payloads, parsed payloads, intent classification, workflow dispatch logs

3. Parsing pipeline missing

- No body cleaning service
- No metadata extractor service

4. Intent engine missing

- No deterministic + LLM hybrid classifier for required four intents
- No confidence threshold/fallback handling

5. Workflow dispatcher missing

- No configurable client hook registry
- No queue publication path for inbound intents
- No retry/dead-letter policy specific to workflow callbacks

6. Inbound lifecycle observability missing

- No inbound metrics (capture rate, parse success, classification accuracy, dispatch success)
- No operator dashboard for reply automation

3.2 Data & Compliance Gaps

- No inbound raw content retention policy implementation
- No PII masking for inbound payload logs
- No immutable inbound audit trail with stage-by-stage trace
- No data subject tooling for inbound data access/erasure in this flow

3.3 Reliability Gaps

- No idempotency strategy for duplicated inbound webhook deliveries
 - No poison message handling/dead-letter queue path for malformed inbound payloads
 - No contract tests for external provider inbound formats
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4. Target-State Architecture for Additional Requirements

Proposed inbound pipeline:

1. Inbound Adapter Layer

- Provider-specific webhook adapters (email, sms/whatsapp, chat)
- Normalize into canonical `InboundMessage`

2. Inbound Processing Layer

- Raw payload persistence
- Parsing/cleaning
- Metadata extraction
- Intent detection + confidence score

3. Workflow Automation Layer

- Rules/route resolver based on intent + tenant config
- Dispatch to:
 - client API webhook
 - or queue topic

4. Observability & Audit Layer

- Inbound event timeline
 - Retry/dead-letter handling
 - Metrics and operator views
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5. Execution Plan (Epics and Tasks)

Epic 1: Inbound Domain & Data Foundation

Goal: Introduce inbound-first schema and domain models.

Tasks

- E1-T1: Create tables: `inbound_messages_raw`, `inbound_messages_parsed`, `inbound_intents`, `inbound_workflow_events`
- E1-T2: Add tenant/user linkage fields (`tenant_id`, `candidate_id`, `candidate_email`, `external_message_id`)
- E1-T3: Add idempotency keys and dedup indexes for inbound payloads
- E1-T4: Add retention metadata columns and archival flags
- E1-T5: Add SQLAlchemy models + Alembic migration + repository layer

Deliverables

- Migration scripts
 - ORM models
 - CRUD/repository abstractions
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Epic 2: Inbound Capture Adapters

Goal: Capture inbound replies from supported channels and normalize payloads.

Tasks

- E2-T1: Define canonical inbound contract (`InboundMessageCanonical`)
- E2-T2: Implement provider adapters for email, SMS/WhatsApp, and chat channels
- E2-T3: Add authenticated inbound webhook endpoints and signature verification per provider
- E2-T4: Persist raw payload + canonical payload atomically
- E2-T5: Add idempotent receive behavior and duplicate suppression

Deliverables

- New `src/api/routers/inbound.py`
 - Adapter services in `src/services/inbound_*`
 - Integration tests with fixture payloads
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Epic 3: Parsing & Entity Extraction

Goal: Parse and clean inbound content to structured form.

Tasks

- E3-T1: Implement message cleaner (signature/thread removal)
- E3-T2: Implement candidate/message metadata extractor

- E3-T3: Support fallback extraction from headers and provider metadata
- E3-T4: Persist parsed result with parser version and confidence flags
- E3-T5: Add parse failure taxonomy + retry-safe error states

Deliverables

- `inbound_parser` service
 - Parsing test corpus and gold outputs
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Epic 4: Intent Detection Engine

Goal: Classify parsed message into required intents.

Tasks

- E4-T1: Define intent taxonomy and decision policy (`accept/reject/request/query`)
- E4-T2: Implement deterministic rules (keyword/regex) baseline
- E4-T3: Implement LLM fallback classifier for ambiguous cases
- E4-T4: Add confidence scoring + threshold + `needs_review` state
- E4-T5: Persist intent result and rationale snapshot for audit

Deliverables

- `intent_service` with deterministic + LLM pipeline
 - Unit/benchmark tests and confusion matrix report
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Epic 5: Workflow Automation Dispatcher

Goal: Trigger downstream automation via API hook or MQ.

Tasks

- E5-T1: Create tenant-configurable intent routing rules
- E5-T2: Implement API webhook dispatcher (signed outbound callback)
- E5-T3: Implement MQ publisher path (Redis/Celery queue first; pluggable for SQS/Kafka)
- E5-T4: Add retry with exponential backoff and dead-letter handling
- E5-T5: Persist dispatch attempts and final outcome states

Deliverables

- `workflow_dispatcher` service
 - Route config model + admin/tenant configuration APIs
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Epic 6: Inbound Audit, Security, and Compliance Controls

Goal: Make inbound pipeline operationally and compliance safe.

Tasks

- E6-T1: Add inbound audit log stream across all stages
- E6-T2: Mask/redact sensitive fields in logs and dashboard views
- E6-T3: Add retention jobs for inbound raw/parsed content
- E6-T4: Add access and erasure APIs for inbound records (tenant scoped)
- E6-T5: Add security event logging for failed signatures/auth attempts

Deliverables

- Compliance-ready inbound audit tables/events
 - Retention and redaction job scripts
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Epic 7: Observability, Dashboard, and Operations

Goal: Operate inbound automation with measurable SLAs.

Tasks

- E7-T1: Define inbound KPIs (capture success, parse success, intent confidence, dispatch success)
- E7-T2: Add dashboard APIs/widgets for inbound pipeline stages
- E7-T3: Add alerting thresholds for failure spikes and dead-letter growth
- E7-T4: Add replay tooling for failed inbound messages
- E7-T5: Add trace IDs across capture -> parse -> intent -> dispatch

Deliverables

- Inbound analytics endpoints
 - Ops runbook and replay tooling
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Epic 8: Testing, Rollout